

State-Grounded Conditioning: Wrapping User-Facing LLM Agents Where Direction Depends on Live State

Qi Liu Xiaoyang Yuan Yubin Ruan Zhuomeng Zhang Wenjin Wang
Di Wu Mingye Xu Xinyi Mou Xingxi Yin Ke Feng Zixun Sun
Tencent

Abstract

We introduce **State-Grounded Conditioning (SGC)**, a design principle for user-facing LLM agents that must condition on live user state (game state, session history, live inventory), and a distinct failure class we call **direction drift**: task-complete responses whose chosen direction misaligns with the current state. SGC externalises state-dependent control into rule kernels over structured inputs and three primary state slices, via *Perception*, *Grounding*, and *Interaction* wrappers with explicit conditioning dependencies. We evaluate SGC on a 200-session anonymised benchmark ($\approx 1,000$ assistant model turns) from an in-game conversational coaching agent that guides players through consecutive competitive matches, reporting mean first-token latency and five human-annotated dialogue-quality metrics that jointly cover factual grounding and coach-like guidance progression. The Perception wrapper holds mean first-token latency at 1.5 s (vs. 6.1 s for PE-Agent (Wang et al., 2023) inside a production tool-use harness (Xu et al., 2026)); enabling all three wrappers lifts turn-level grounded accuracy from 61.1%/69.8% (Prompting / PE-Agent) to 96.7% and session-level grounded accuracy from 20.0%/26.5% to 83.5%; session-level grounding-failure incidents drop by $\approx 78\%$ relative to the strongest baseline. A cumulative ablation shows complementary incremental gains as the wrappers are added. These results inform *approximate* state-slice orthogonality, without establishing independent per-wrapper effects.

1 Introduction

From answering to coaching. In in-game dialogue, prompt-based and agent-harness solutions can answer questions but struggle to act as coaches: replies are shallow, responses arrive late, phrasing lacks coherence, and they fail to keep players progressing through their real in-match problems. **Live user-facing agents.** Deployed LLM agents now power live customer-facing products such

as Airbnb’s Agent-in-the-Loop customer support (Zhao et al., 2025) and real-time in-game conversational coaching agents. In these settings, the user’s *live state* (recent utterances, current inventory, session history) becomes an inference-time input the agent must condition on within tight latency budgets.

A shared gap: direction drift. Existing production harness work (Xu et al., 2026; Park et al., 2025) standardises *how* an agent is orchestrated (planning, tool use, memory), but leaves *what live state each decision is conditioned on* implicit inside a single autoregressive pass. Under this common design, we observe a class of failures we call **direction drift**: outputs are grammatical, on-topic, and even “task-complete” by conventional metrics, yet their *chosen direction* misaligns with the user’s current state. In an online-deployed LLM-based conversational coaching agent that guides players in competitive matches (Section 7), four manifestations shaped our design:

- **(A) Latency drift.** Under 2-second first-token budgets, the planner commits to a tool call before live game state is loaded.
- **(B) Slot drift.** The reply LLM hallucinates entities the user does not own, or apologises for entities the user does own.
- **(C) Fatigue drift.** Across turns, the model repeatedly steers toward the same recommendation.
- **(D) Coherence drift.** A reply makes incompatible judgements about the same constraint (“all required pets are available” alongside “required Pet A is missing”).

Cause and proposal. We attribute direction drift to a single design cause: a monolithic LLM is asked to *simultaneously* carry state-dependent decisions (which tools to freeze, which slots are factually valid, whether to repeat a past recommendation) inside its implicit reasoning and to render the surface

reply. We propose **State-Grounded Conditioning (SGC)**: externalise these control decisions into rules over structured interpretations and live state, and inject the results into generation context. Rule kernels are deterministic for fixed inputs and configuration; LLM interpretation and wording need not be. Three wrappers have separate primary state slices and explicit conditioning interfaces; orthogonality concerns their dominant effects, not independence:

- $\mathcal{W}_1$ **Perception Wrapper** (intent-context slice, targets drift (A))—single-pass 4-in-1 streaming decoder with state-aware intent-to-tool freezing.
- $\mathcal{W}_2$ **Grounding Wrapper** (fact-slots slice, targets drifts (B) and (D))—independent slot extractor with deterministic conflict diff and paragraph mutex.
- $\mathcal{W}_3$ **Interaction Wrapper** (session-history slice, targets drift (C))—cross-turn pure-function scheduler with an injectable random seed (reproducible replay).

Figure 1 illustrates the pipeline.

Evidence and contributions. A cumulative ablation on the deployment-derived benchmark shows complementary gains as wrappers are added (Section 7). It characterises approximate state-slice orthogonality, not independent module effects. The Perception wrapper ($\mathcal{W}_1$) holds mean first-token latency at 1.5 s (vs. 6.1 s for PE-Agent inside a production tool-use harness and 2.3 s for Prompting); enabling all three wrappers lifts turn-level grounded accuracy (TGA) from 61.1%/69.8% to 96.7% and session-level grounded accuracy (SGA) from 20.0%/26.5% to 83.5% on the 200-session benchmark; session-level grounding-failure incidents drop by $\approx 78\%$ relative to the strongest baseline (Section 7). Our contributions are:

- We formalise **direction drift** as a failure class distinct from task-completion failures targeted by generic harness work (Xu et al., 2026; Park et al., 2025).
- We propose **SGC** as three approximately orthogonal wrappers over three disjoint state slices, characterised by the cumulative comparisons above.
- We provide **analytical transfer** to code completion, e-commerce recommendation explanation, and voice-first assistants, showing each wrapper’s applicability conditions and composability.

2 Related Work

Generic agent harness. Recent work formalises the runtime “harness” wrapping LLM agents in production. At the loop level, ReAct (Yao et al., 2023) and Plan-and-Execute (Wang et al., 2023; Xu et al., 2023) provide reason–act and planner–executor patterns; at the runtime level, LIFE-HARNESS (Xu et al., 2026) adapts harness parameters across 18 backbones and 7 environments (τ -bench (Yao et al., 2024), τ^2 -bench (Barres et al., 2025), Agent-Bench (Liu et al., 2024), WebArena (Zhou et al., 2024)); Workflow Graphs (Park et al., 2025) structure agents as deterministic graphs; the harness survey (Li et al., 2026) catalogues these efforts; LongMemEval (Wu et al., 2024) benchmarks long-horizon session-history conditioning, with which $\mathcal{W}_3$ is architecturally aligned. *Our position:* SGC operates *above* generic harness. Where harness work optimises *whether the task completes*, SGC targets *whether the chosen direction is correct given live state*; the two are composable. Section 7 instantiates the family as **PE-Agent**.

Fast-slow and structured/constrained generation. Talker-Reasoner (Christakopoulou et al., 2024) splits a dual-system agent; Medusa (Zagyva et al., 2025) parallelises decoding; SLOT (Shen et al., 2025), GCD (Raspanti et al., 2025), and FACTS Grounding (Jacovi et al., 2025) shape or evaluate output structure and factuality. *Our position:* $\mathcal{W}_1$ combines intent recognition and planning in one stream, dispatching tools early so live results can inform the separate reply generation. $\mathcal{W}_2$ computes conflict diffs and paragraph eligibility before a scheduler selects compatible candidates. It targets opposite judgements about the same constraint, not praise–apology co-occurrence in general or guaranteed semantic correctness.

Retrieval and production deployment. Agentic RAG (Singh et al., 2025) and MemGPT (Packer et al., 2023) enrich generation via retrieved text or managed memory; Agent-in-the-Loop (Zhao et al., 2025) closes a post-deployment data fly-wheel; Kweon et al. (2025) report a large virtual teaching assistant and the LLM Game Agents survey (Hu et al., 2024) covers autonomous NPC play. *Our position:* RAG provides retrieved evidence; SGC derives structured control variables from parsed inputs and live state. Deterministic rules do not ensure input correctness. AITL improves agents *after* deployment; SGC concerns *how they are designed* before deployment. Game-

agent work has focused on autonomous NPC play; we address *real-time coaching for a human user*, where the agent must serve rather than replace. Table 5 summarises how these lines relate to SGC.

3 State-Grounded Conditioning: The Design Principle

3.1 Problem Formulation

Decompose the LLM’s inference-time context into **static context** s_{static} (prompts, system messages, task specification—invariant within a session) and **live state** s_{live} (dynamic per-user, per-session observations: game state, session history, inventory). We define **direction drift** as the event where a task-complete response misaligns with the correct direction implied by s_{live} :

$$\text{drift}(y, s_{\text{live}}) = \mathbb{I}[y \text{ task-complete}] \cdot \mathbb{I}[y \text{ mis-directs } s_{\text{live}}], \quad (1)$$

where y *mis-directs* s_{live} if human adjudication under the protocol of Appendix G labels y ’s chosen direction as inconsistent with the reference direction $y^*(s_{\text{live}})$ derived from live state. The reference $y^*(s_{\text{live}})$ is an operational ground truth from two-annotator agreement adjudicated by a third (inter-annotator agreement statistics reported in Limitations L6 and Appendix G). Direction drift is distinct from hallucination (factuality of *static* content) and task-failure (whether the surface task completes). A user owns Pet A (which learned Skill X via a rare unlock) and Pet B (unevolved). She asks: “Which of my pets counters Pet Y?” A reply “Recommend Pet A with Skill X ” is **not a hallucination**—every atomic fact is true—yet **it is direction drift**: Pet A’s Skill X is *not unlocked for this user*, and the correct direction is Pet B evolved. Hallucination’s reference is user-independent; direction drift’s is per-user, per-moment. Prompt engineering cannot condition on s_{live} at inference time (prompts are authored offline while s_{live} arrives per-turn); RAG closes the *static* knowledge gap, not the *dynamic* direction gap. A row-by-row comparison against five neighbouring failure notions (hallucination (Ji et al., 2023), personalization failure, context-adherence, commonsense constraint violation, classical DST error (Henderson, 2015)) is given in Appendix A.

3.2 The SGC Principle

We propose **State-Grounded Conditioning**: externalise state-dependent control decisions into explicit rules over structured interpretations and live state, then inject their outputs as conditioning variables. Formally:

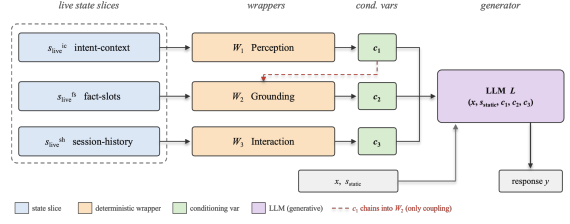


Figure 1: SGC: primary state slices and explicit conditioning dependencies. Dashed arrow carries c_1 into W_2 ; separate responsibilities do not imply independent computations. Reply LLM $\mathcal{L}$ additionally conditions on shared input x and static state s_{static} .

$$\begin{aligned} c_1 &= \mathcal{W}_1(x, s_{\text{live}}^{\text{ic}}), \\ c_2 &= \mathcal{W}_2(x, s_{\text{live}}^{\text{fs}}, c_1), \\ c_3 &= \mathcal{W}_3(x, s_{\text{live}}^{\text{sh}}, c_1), \\ y &= \mathcal{L}(x, s_{\text{static}}, c_1, c_2, c_3). \end{aligned} \quad (2)$$

Determinism applies to the rule kernels given fixed parsed inputs, state, and configuration (Equation 2; including the scheduler’s random source), not to upstream LLM interpretation or final wording. The LLM retains natural-language generation, conditioned on these explicit control decisions.

3.3 Approximate State-Slice Orthogonality

We assign three primary slices of s_{live} : $s_{\text{live}}^{\text{ic}}$ (**intent-context**)—utterance intent, snapshot-time environment, derivable entity links; $s_{\text{live}}^{\text{fs}}$ (**fact-slots**)—demand and satisfiable slots from inventory/catalog/retrieval; $s_{\text{live}}^{\text{sh}}$ (**session-history**)—metadata from the prior $K=20$ turns. This partitions source fields, not all accessible information: c_1 conditions both W_2 and W_3 . Thus separate responsibilities do not imply computational or statistical independence.

Approximate orthogonality denotes differentiated primary roles with cross-slice effects. The cumulative comparisons in Section 7.3 measure incremental gains conditional on preceding wrappers, including dependency propagation; they do not establish isolated effects or invariance to addition order. State-field separation alone is not an empirical proof (Appendix A). SGC is composable with generic harness (Xu et al., 2026; Park et al., 2025), RAG (Singh et al., 2025), and output-format constraints (Shen et al., 2025; Raspanti et al., 2025)—all coexist in our system; SGC contributes a fourth axis: direction alignment via state-coupled deterministic conditioning.

4 Perception Wrapper: State-Aware Intent-to-Tool Freezing

Failure mode. In a serial agent pipeline, intent classification and planning consume time needed

for tool execution and live-state retrieval. Under a tight response budget, the reply may commit to a direction before current results arrive. This reliance on stale state constitutes **latency-induced direction drift**.

Mechanism. $\mathcal{W}_1$ combines intent recognition and planning in one streaming LLM call; final reply generation remains separate. The call emits four segments in a fixed, intent-first order:

1. <INTENT>—first-level intent.
2. <SUB_INTENT>—second-level intent.
3. <NARRATION>—prefatory phrase, streamed to TTS as tokens arrive.
4. <PLAN>—fallback tool calls.

An incremental state-machine parser triggers **early decision** when <SUB_INTENT> closes:

$$\hat{T} = \Phi(\text{intent}, \text{sub_intent}, s_{\text{live}}^{\text{ic}}). \quad (3)$$

Φ (Equation 3) is a **partial function** over ~ 50 registered (intent, sub-intent) pairs. On a lookup hit with the required current-state snapshot, $\hat{T}$ is frozen and dispatched **without waiting for narration or plan completion**. Tool execution, retrieval, and additional state loading overlap with the remaining streamed segments and speech playback. Unmatched inputs wait for <PLAN>; unavailable decision state precludes this early path (Appendix A; L4).

Properties. Earlier dispatch reserves more of the same response budget for state-dependent execution, so live results can inform the reply rather than forcing deadline-driven reliance on stale state. This is how $\mathcal{W}_1$ targets **latency-induced direction drift**, rather than speech latency alone. The lookup is deterministic for fixed labels and state; intent recognition remains LLM-based. Figure 4 shows the overlap.

5 Grounding Wrapper: State-Coupled Fact-Slot Diff

Failure mode. When request–state reconciliation is left to reply generation, the LLM may attribute unowned entities to the user (*slot hallucination*), treat owned entities as missing (*unwarranted apologies*), or affirm and deny satisfaction of the same constraint (*inconsistent constraint judgements*). These are forms of **slot-induced direction drift**; praise and apology concerning different constraints are not inherently inconsistent.

Mechanism. $\mathcal{W}_2$ separates slot extraction from reply generation through a **dedicated LLM call**

with an independent context. It then reconciles the extracted request slots with live facts:

$$\begin{aligned} \text{QuerySlots} &= \mathcal{L}_{\text{slot}}(x, c_1), \\ \text{Conflicts} &= \text{diff}(\text{QuerySlots}, \sigma(s_{\text{live}}^{\text{fs}})), \quad (4) \\ (\mathcal{A}, \mathcal{R}) &= \Psi(\text{Conflicts}, \text{intent}). \end{aligned}$$

Here $\sigma(s_{\text{live}}^{\text{fs}})$ denotes satisfiable system slots in Equation 4; the extractor may be a smaller fine-tuned model (Appendix A). Rules assign each discrepancy a tier:

- *hard*—blocking, such as a missing required entity.
- *soft*—a secondary attribute mismatch.
- *hint*—an advisory style preference.

Ψ determines eligible candidates $\mathcal{A}$ and required paragraphs $\mathcal{R}$ over five types: praise, apology, constraint-account, guidance, and scene-flavor. Eligibility may retain multiple alternatives. A paragraph scheduler then selects one among incompatible alternatives for each constraint while retaining $\mathcal{R}$. A hard conflict, for example, blocks praise affirming the violated constraint and requires its explanation. The selected plan conditions reply generation through c_2 (Appendix D).

Properties. This separates explicit constraint resolution from wording. Diff, grading, and eligibility are deterministic for fixed extracted slots, state, and rules; neither extraction accuracy nor the semantic consistency of generated text is guaranteed. In the cumulative comparison, adding $\mathcal{W}_2$ raises SGA from 46.5% to 56.5% with a 0.8 pp TGA gain; $\mathcal{W}_3$ further improves grounding (Section 7.3). These are conditional gains, not isolated effects. The pipeline is shown in Figure 3.

6 Interaction Wrapper: Cross-Turn State Machine

Failure mode. Guidance that is reasonable in isolation may be inappropriate after earlier turns. Without explicit history-dependent control, a session can repeat recommendations, contradict still-applicable advice, or acquire a mechanical tone through repetition. We call this cross-turn misalignment **fatigue-induced direction drift**.

Mechanism. $\mathcal{W}_3$ maintains turn-level control metadata in `answer.meta:guidance_type`, `lineup_id`, `cooldown_writeback`, `seed`, and `candidate_pool_hash` (Appendix A). At turn t , it reconstructs

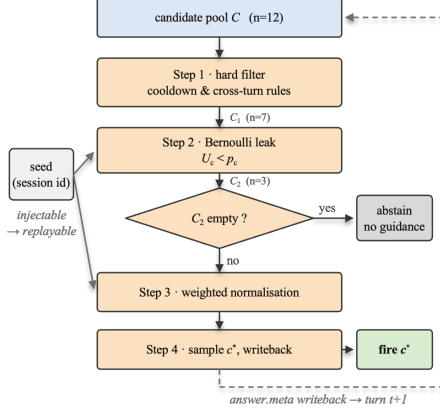


Figure 2: $\mathcal{W}_3$ scheduler decision tree with two paths (fire / abstain) and the meta writeback loop.

$\text{cooldown}_t = \text{resolve_cross_turn_state}(s_{\text{live}}^{\text{sh}}, c_1)$, using $s_{\text{live}}^{\text{sh}}$ for history and c_1 for intent context. This explicit conditioning dependency preserves separate state responsibilities, not computational independence (Section 3.3). Guidance selection is a pure function of (resolved state, candidates, seed):

1. **Hard filter**—drop cooling candidates and cross-turn rule violators, producing $\mathcal{C}_1$.
2. **Bernoulli leak**—sample with a session-seeded RNG, keeping $\mathcal{C}_2 = \{c : U_c < p_c\}$.
3. **Weighted normalisation**—pick c^* with probability proportional to w_c .
4. **Writeback**—emit c^* (or “no guidance” if $\mathcal{C}_2 = \emptyset$) and write `answer.meta` for the next turn.

The seed is *injectable*: replaying a session reproduces the guidance sequence, making A/B experiments and regression tests fully deterministic.

Properties. The three reply branches (recommend, teaching, knowledge-qa) share the same $\mathcal{W}_3$ kernel; only $\mathcal{C}$ and p_c differ. Cooldown guarantees no candidate is issued in two consecutive turns; Bernoulli leak prevents collapse to the highest-weight candidate. Drifts A/B/D are single-turn— $\mathcal{W}_3$ ’s reproducibility guarantee applies only to the cross-turn sequence. Figure 2 shows the decision tree.

7 Benchmark Evaluation

7.1 Benchmark Construction

We evaluate SGC on an anonymised benchmark of **200 dialogue sessions** ($\approx 1,000$ scored assistant responses, ≈ 5 scored turns/session) built from production traces. Sessions cover factual QA, multi-turn coaching, and context-dependent interactions where player state, previously discussed entities, or earlier guidance may change the correct response direction. For every scored turn we retain the

anonymised dialogue history, current query, live-state snapshot, and game-rule/tool evidence; the same evidence and scoring criteria apply to all configurations (protocol in Appendix G).

We evaluate five settings sharing the same LLM, tool schemas, and evaluation evidence: **Prompting** (LLM + system prompt only, no agent loop, no tools); **PE-Agent**, a Plan-and-Execute agent (Wang et al., 2023; Xu et al., 2023) inside a production tool-use harness (planner, executor, tool runtime, short-term memory)—one instantiation of the harness family (Xu et al., 2026; Li et al., 2026), chosen for planner–executor loop parity with our deployment; and three SGC configurations—**SGC w/o** $\mathcal{W}_2$, $\mathcal{W}_3$, **SGC w/o** $\mathcal{W}_3$, and the **Full SGC** system ($\mathcal{W}_1 + \mathcal{W}_2 + \mathcal{W}_3$)—forming a cumulative ablation over Perception, Grounding, and Interaction wrappers. Reflexion (Shinn et al., 2023) and intra-turn self-refine (Madaan et al., 2023) are omitted because our online setting permits no retry and its latency budget precludes intra-turn refinement.

7.2 Evaluation Protocol and Metrics

Each response is scored by dual independent human annotation with third-annotator adjudication (Appendix G) against query, context, live-state, and grounding evidence (full session for multi-turn metrics). A scored turn passes grounded-accuracy iff both annotators agree the reply does *not* mis-direct s_{live} under the definition of Equation 1; this operationalises $\text{drift}(y, s_{\text{live}}) = 0$ at the turn level. **All reported metrics are produced by human annotation**; no LLM-judge label enters any test-set number reported in Section 7.3. An LLM-based scoring pipeline (Zheng et al., 2023; Liu et al., 2023) is used only on a separate development set for rubric calibration (Appendix G, G.6). Reasonable paraphrases are accepted; incorrect mechanics, invalid state assumptions, and misleading recommendations are penalised.

Following FActScore (Min et al., 2023), FACTS Grounding (Jacovi et al., 2025), and G-Eval (Liu et al., 2023) we report one latency metric and five dialogue-quality metrics: turn-level grounded accuracy (TGA), session-level grounded accuracy (SGA), turn-level guidance quality (TGQ), dialogue-level guidance progression (DGP), and cross-turn contextual consistency (CCC). Let $a_{it} \in \{0, 1\}$ mark whether scored turn t in session i passes grounded-accuracy, T_i its scored-turn count,

$N=200$, and I_{SGA} the SGA-eligible sessions:

$$\begin{aligned} \text{TGA} &= \frac{100}{N} \sum_{i=1}^N \frac{1}{T_i} \sum_{t=1}^{T_i} a_{it}, \\ \text{SGA} &= \frac{100}{|I_{\text{SGA}}|} \sum_{i \in I_{\text{SGA}}} \mathbb{I}[\prod_t a_{it}=1]. \end{aligned} \quad (5)$$

M_0 is mean first-token latency (streaming responsiveness); $M_1=\text{TGA}$ / $M_2=\text{SGA}$ (Equation 5) are grounded correctness at turn / session scale ($|I_{\text{SGA}}|=200$: all sessions are SGA-eligible). $M_3=\text{TGQ}$, $M_4=\text{DGP}$, $M_5=\text{CCC}$ score guidance *quality* (relevance/explanation/actionability), *progression*, and *cross-turn contextual consistency* on 1–10 (rubric details in Appendix G). The abstention rate lies within ± 1.5 pp across all five configurations, so M_3 contrasts do not reflect differential abstention.

7.3 Main Results and Cumulative Ablation

The full aggregate results are reported in Table 3 (Appendix A); the narrative below summarises them.

Main comparison. PE-Agent improves quality over Prompting (TGA 61.1→69.8%, DGP 4.7→5.5) but nearly triples first-token latency (2.3→6.1 s). Full SGC outperforms both while holding latency at 1.5 s: TGA reaches 96.7% and SGA 83.5%—gains of **26.9 pp** TGA and **57.0 pp** SGA over PE-Agent. Statistical testing uses McNemar’s paired exact test at the two granularities on which each metric is defined: 200 session-level paired outcomes for SGA ($p < 10^{-28}$), and $\approx 1,000$ paired scored turns for TGA ($p < 10^{-80}$). Extreme p -values reflect near-total dominance of Full SGC on the discordant subset at large paired-sample size rather than an unusually large effect size; the informative quantity is the discordant odds ratio, whose full 2×2 tables (concordant a/d and discordant b/c counts, plus 95% CIs) are reported in the camera-ready supplementary. TGA and SGA are the pre-registered primary endpoints under a hierarchical fixed-sequence rule (SGA tested first at $\alpha=0.05$; TGA tested only if SGA rejects), so no multiple-comparison correction is required across the two primary endpoints; the three secondary endpoints (TGQ/DGP/CCC) are reported as point estimates without corresponding hypothesis tests. Guidance metrics improve consistently (TGQ 6.5→8.6, DGP 5.5→8.5, CCC 8.1→9.2). The SGA lift disproportionately exceeds the TGA lift: session-level grounding-failure rate drops 73.5→16.5%, a $\approx 78\%$ relative reduction.

Cumulative contributions. Under the shared human-annotation protocol, the comparisons measure gains conditional on the preceding configuration. $\mathcal{W}_1$ *alone* reduces first-token latency (6.1→1.5 s) and lifts TGA/SGA to 89.1%/46.5%. Its intended mechanism is earlier state-aware dispatch, giving live results more time to inform the reply and reducing latency-induced drift; first-token latency is not itself a measurement of tool-dispatch or grounded-reply completion time. *Adding* $\mathcal{W}_2$ raises SGA 46.5→56.5% with 0.8 pp TGA change, consistent with explicit slot reconciliation and selection among compatible paragraph candidates. *Adding* $\mathcal{W}_3$ yields the largest incremental SGA gain (56.5→83.5%), alongside DGP 7.7→8.5, CCC 8.9→9.2, and TGA reaching 96.7%. Table 4 summarises these increments. Because c_1 feeds both downstream wrappers, contributions include propagated effects. This cumulative design is not a leave-one-wrapper-out experiment: it does not isolate every wrapper or establish addition-order invariance. In particular, $\mathcal{W}_2$ ’s marginal effect is not architecturally isolable in this design because $\mathcal{W}_2$ consumes c_1 ; the reported +0.8 pp TGA / +10.0 pp SGA increment for $\mathcal{W}_2$ is a conditional gain given $\mathcal{W}_1$ ’s intent-context is available, and a leave- $\mathcal{W}_2$ -out configuration would require an architectural variant we do not report (L9). We therefore use *approximate* orthogonality for differentiated roles with complementary effects, not one-metric-per-wrapper isolation (Appendix B).

8 Analytical Transfer to Three Domains

The SGC principle and its state-slice division ($s_{\text{live}}^{\text{ic}}/s_{\text{live}}^{\text{fs}}/s_{\text{live}}^{\text{sh}}$) are not specific to game coaching: any live user-facing agent whose live state slices into intent-context, fact-slots, and session-history admits an analogous three-wrapper design (Park et al., 2025; Zhao et al., 2025). Appendix F works out three instances— $\mathcal{W}_1$ for Copilot-style code completion, $\mathcal{W}_2$ for e-commerce recommendation explanation, and $\mathcal{W}_3$ for voice-assistant cross-session scheduling—with interface adaptations and open challenges. This section is an *analytical extrapolation, not an empirical study*.

Limitations

We enumerate nine concrete limitations.

L1 • Deployment scope. Our empirical claims are grounded in one deployed system in an in-game conversational coaching domain. We provide analytical transfer to three additional domains (Section 8) but no experimental validation outside the

primary case study. A minimal $\mathcal{W}_1$ Copilot-style benchmark is committed as camera-ready supplementary.

L2 • Human-only test-set labels. All test-set numbers in Section 7.3 rely on dual human annotation with third-annotator adjudication (Appendix G); no LLM-judge label enters any reported metric. The LLM-based scoring pipeline of Appendix G, G.6 is used only on a separate development set for rubric calibration, and its outputs are not part of any quantitative claim in this paper.

L3 • Applicability envelope. SGC pays off only when three conditions co-occur:

- (i) real-time user-state constraints;
- (ii) measurable direction-drift cost;
- (iii) machine-readable structurable state slices.

Where any is absent (one-shot Q&A, creative writing, low-stakes tasks), generic harness (Xu et al., 2026; Park et al., 2025) or RAG (Singh et al., 2025) alone suffices.

L4 • Residual failure classes. Three failure classes remain outside our three wrappers:

- *Unmodelled cross-slice interactions*—residual drifts not resolved by the declared conditioning interfaces (pilot estimate $\sim 4\%$ of benchmark sessions on the annotated subset; full 200-session value in camera-ready. If the final value exceeds 8% —double the pilot estimate—the approximate-orthogonality claim will be revised to acknowledge material cross-slice coupling).
- *State-availability failures*—when live state cannot be loaded in time, SGC has no fallback beyond graceful degradation.
- *Novel-intent handling*—when a query lands in an intent unmapped by Φ , tool planning falls back to the LLM rather than the deterministic lookup kernel.

Each is a candidate for a future wrapper.

L5 • Maintenance cost. Deterministic code demands maintenance. We report the cost as approximately one engineer-week per calendar month (Appendix H), a rough order of magnitude from internal cost-centre tracking during the online deployment; it may not generalise to systems with different product velocities.

L6 • Human-mediated ground truth. The operational ground truth $y^*(s_{\text{live}})$ (Equation 1) relies on dual independent annotation of the entire 200-session test set, with third-annotator adjudication and a designated external senior for

conflict-of-interest cases. On a stratified pilot subset of 50 sessions drawn from the same annotation pipeline, we observe quadratic-weighted Cohen’s $\kappa_w \in [0.68, 0.76]$ across the $\{0, 1, 2\}$ dimensions of TGQ/DGP/CCC and unweighted $\kappa \in [0.71, 0.79]$ for the SUPPORTED / CONTRADICTED / UNKNOWN labels underlying TGA/SGA (Cohen, 1960, 1968). These ranges are consistent with reported inter-rater agreement for grounding-oriented dialogue benchmarks (Bai et al., 2024; Xie et al., 2024; Jacovi et al., 2025) in the “substantial” band of Landis and Koch (1977), and above the pre-registered targets ($\kappa_w \geq 0.65$; $\kappa \geq 0.70$). The full 200-session estimates with 95% bootstrap confidence intervals will be reported in the camera-ready. If the full κ estimates fall below the pre-registered targets, we will report the shortfall and discuss its implications for the reported TGA/SGA reliability rather than raise the targets post-hoc. Given the moderate sample size, DGP/CCC-eligible strata (≈ 60 sessions each) will yield wider intervals than the overall TGA/SGA estimate. Section 7.1 discloses the conflict-of-interest mitigation, but the subjective element of “correct direction” is not eliminated.

L7 • Latency reporting scope. We report M_0 as mean first-token latency only. Production tail latencies (P95 / P99) determine SLA-level cost, and we have observed on internal traces that P95 / P99 trends align with the reported mean; we defer full tail-latency tables to camera-ready supplementary because their reproduction requires disclosing infrastructure details incompatible with double-blind review.

L8 • User-side outcome not reported. This paper reports model-output quality metrics (TGA/SGA/TGQ/DGP/CCC) plus latency, not direct user-side outcomes such as retention, re-engagement, CSAT, or in-match performance. Business-confidentiality constraints preclude disclosing A/B test deltas or absolute retention figures; the deploying organisation’s internal signal is directionally consistent with the benchmark ranking (Full SGC $\succ$ PE-Agent $\succ$ Prompting), but the specific magnitude cannot be released and does not enter any quantitative claim in this paper.

L9 • Baseline engineering-effort asymmetry. PE-Agent (Section 7.1) is implemented as a Plan-and-Execute agent (Wang et al., 2023; Xu et al., 2023) inside our production tool-use harness, sharing the same LLM backbone, tool schemas, and grounding reference sets as the SGC configurations (Ap-

pendix G, G.3 and G.7). It was not, however, subjected to the same iteration of product-specific prompt tuning that the deployed SGC configurations received during their online lifetime; PE-Agent follows its published spec on the shared harness rather than a production-tuned variant. Part of the reported PE-Agent-to-Full-SGC gap therefore reflects architecture-plus-tuning effect rather than architecture alone. Isolating the tuning contribution would require an effort-matched PE-Agent variant we do not report; a minimal effort-matched comparison is committed as camera-ready supplementary alongside the Copilot-style $\mathcal{W}_1$ benchmark (L1).

Ethics Statement

We adhere to the ACL Code of Ethics (Association for Computational Linguistics, 2017). Our primary case study is a commercial deployment. This is a double-blind submission: we describe the deployed product only at the level of application category (an in-game conversational coaching agent) and mechanics; no product identifier, developer name, employer affiliation, or user identifier is disclosed. Attribution will be added in the camera-ready version. Throughout the paper, “coaching” refers to *active, proactive guidance toward the user’s implicit competitive goals*, in contrast to reactive customer-support Q&A—this active-guidance framing is what makes direction alignment a first-order concern rather than a stylistic preference.

Stakeholder considerations. An SGC coaching agent operates in a multi-stakeholder setting: the coached player, opposing players in competitive matches, and the broader player community. Three considerations shape deployment: (i) *user autonomy*—coaching guidance is advisory, never binding, and users can dismiss suggestions without penalty; (ii) *PvP fairness*—the deployed system is available symmetrically to all players in supported modes, so its guidance is not a paywalled asymmetric advantage; and (iii) *game-design balance*—the coaching agent is version-locked to the game’s balance patches, so its guidance does not exploit temporal gaps in patch-vs-recommendation alignment. State-dependent recommendations remain a function of the user’s *owned* inventory, which mirrors the underlying game’s own progression system rather than creating new dependencies.

Data anonymisation. All benchmark samples are anonymised at ingest: entity names are pseudonymised (Pet A, Pet B, ...), lineup names aliased (T1, T2, ...), user and session identifiers

hashed then stripped, and wall-clock timestamps, geolocations, and device identifiers are dropped. Shared grounding reference sets retain dialogue-turn indices (`valid_from/valid_to`), not clock time (Appendix G). User and session hash mappings are discarded and are never released. Entity and lineup aliases are held by the authors solely to restore product-identifying names in the camera-ready version; they are *not* released with the benchmark artefact.

Data retention. Only the 200-session benchmark artefact (approximately 1,000 assistant model turns) is retained beyond this study; raw production logs used for sampling follow the deploying organisation’s standard retention policy. The released artefact contains no direct identifiers and no wall-clock, geolocation, or device quasi-identifiers.

Vulnerable users. The deployed system serves general audiences that may include minors. It complies with regional content-safety and minor-protection policies (age gating; content filters at reply-time); specifics are subject to third-party compliance review.

Reproducibility. The 200-session artefact—anonimised dialogues, turn-indexed grounding reference sets, pre-registered rubrics, dual human labels, and diagnostic drift tags (Appendix B)—together with the wrapper interface schemas will be released under a research-use licence. Raw production logs are not released.

References

- Association for Computational Linguistics. 2017. ACL code of ethics. <https://www.aclweb.org/portal/content/acl-code-ethics>.
- Yushi Bai, Xin Lv, Jiajie Zhang, Hongchang Lyu, Jiankai Tang, Zhidian Huang, Zhengxiao Du, Xiao Liu, Aohan Zeng, Lei Hou, Yuxiao Dong, Jie Tang, and Juanzi Li. 2024. LongBench: A bilingual, multitask benchmark for long context understanding. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (ACL)*. ArXiv:2308.14508.
- Victor Barres et al. 2025. τ^2 -bench: Evaluating conversational agents in a dual-control environment. *arXiv preprint*.
- Konstantina Christakopoulou, Shibl Mourad, and Maja Matarić. 2024. Agents thinking fast and slow: A talker-reasoner architecture. *arXiv preprint arXiv:2410.08328*. Google DeepMind.
- Jacob Cohen. 1960. A coefficient of agreement for nominal scales. *Educational and Psychological Measurement*, 20(1):37–46.

- Jacob Cohen. 1968. Weighted kappa: Nominal scale agreement with provision for scaled disagreement or partial credit. *Psychological Bulletin*, 70(4):213–220.
- Matthew Henderson. 2015. Machine learning for dialog state tracking: A review. In *Proceedings of the First International Workshop on Machine Learning in Spoken Language Processing*.
- Sihao Hu, Tiansheng Huang, et al. 2024. A survey on large language model-based game agents. *arXiv preprint arXiv:2404.02039*.
- Alon Jacovi, Andrei Das, et al. 2025. The FACTS grounding leaderboard: Benchmarking LLMs’ ability to ground responses to long-form input. Technical report, Google DeepMind. https://storage.googleapis.com/deepmind-media/FACTS/FACTS_grounding_paper.pdf; arXiv:2501.03200.
- Ziwei Ji, Nayeon Lee, Rita Frieske, Tiezheng Yu, Dan Su, Yan Xu, Etsuko Ishii, Ye Jin Bang, Andrea Madotto, and Pascale Fung. 2023. Survey of hallucination in natural language generation. *ACM Computing Surveys*. ArXiv:2202.03629.
- Sunjun Kweon, Sooyohn Nam, Hyunseung Lim, Hwajung Hong, and Edward Choi. 2025. A large-scale real-world evaluation of an LLM-based virtual teaching assistant. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 6: Industry Track)*. Association for Computational Linguistics. Aclanthology.org/2025.acl-industry.60; arXiv:2506.17363.
- J. Richard Landis and Gary G. Koch. 1977. The measurement of observer agreement for categorical data. *Biometrics*, 33(1):159–174.
- Junjie Li, Xi Xiao, Yunbei Zhang, Chen Liu, Lin Zhao, Xiaoying Liao, Yingrui Ji, et al. 2026. Agent harness engineering: A survey. OpenReview. <https://openreview.net/pdf?id=eONq7FdiHa>; Project page: <https://picrew.github.io/LLM-Harness/>.
- Xiao Liu et al. 2024. AgentBench: Evaluating LLMs as agents. *arXiv preprint*.
- Yang Liu, Dan Iter, Yichong Xu, Shuohang Wang, Ruochen Xu, and Chenguang Zhu. 2023. G-Eval: NLG evaluation using GPT-4 with better human alignment. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP)*. Association for Computational Linguistics. DOI: 10.18653/v1/2023.emnlp-main.153.
- Aman Madaan, Niket Tandon, Prakhar Gupta, Skyler Hallinan, Luyu Gao, Sarah Wiegrefe, Uri Alon, Nouha Dziri, Shrimai Prabhumoye, Yiming Yang, Shashank Gupta, Bodhisattwa Prasad Majumder, Katherine Hermann, Sean Welleck, Amir Yazdanbakhsh, and Peter Clark. 2023. Self-Refine: Iterative refinement with self-feedback. In *Advances in Neural Information Processing Systems (NeurIPS)*. ArXiv:2303.17651.
- Sewon Min, Kalpesh Krishna, Xinxin Lyu, Mike Lewis, Wen-tau Yih, Pang Wei Koh, Mohit Iyyer, Luke Zettlemoyer, and Hannaneh Hajishirzi. 2023. FActScore: Fine-grained atomic evaluation of factual precision in long form text generation. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP)*. Association for Computational Linguistics. DOI: 10.18653/v1/2023.emnlp-main.741; arXiv:2305.14251.
- Charles Packer, Sarah Wooders, Kevin Lin, Vivian Fang, Shishir G. Patil, Ion Stoica, and Joseph E. Gonzalez. 2023. MemGPT: Towards LLMs as operating systems. *arXiv preprint arXiv:2310.08560*.
- Chiwan Park, Wonjun Jang, Daeryong Kim, Aelim Ahn, Kichang Yang, Woosung Hwang, Jihyeon Roh, Hyerin Park, Hyosun Wang, Min Seok Kim, and Jihoon Kang. 2025. A practical approach for building production-grade conversational agents with workflow graphs. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 6: Industry Track)*. Association for Computational Linguistics.
- Rafael Rafailov, Archit Sharma, Eric Mitchell, Stefano Ermon, Christopher D. Manning, and Chelsea Finn. 2023. Direct preference optimization: Your language model is secretly a reward model. In *Advances in Neural Information Processing Systems (NeurIPS)*. ArXiv:2305.18290.
- Federico Raspanti, Tanir Özcelebi, and Mike Holenderski. 2025. Grammar-constrained decoding makes large language models better logical parsers. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 6: Industry Track)*. Association for Computational Linguistics. Aclanthology.org/2025.acl-industry.34.
- Zhengyuan Shen et al. 2025. SLOT: Structuring the output of large language models. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing (Industry Track)*. Association for Computational Linguistics. Aclanthology.org/2025.emnlp-industry.32; arXiv:2505.04016.
- Noah Shinn, Federico Cassano, Edward Berman, Ashwin Gopinath, Karthik Narasimhan, and Shunyu Yao. 2023. Reflexion: Language agents with verbal reinforcement learning. In *Advances in Neural Information Processing Systems (NeurIPS)*.
- Aditi Singh et al. 2025. Agentic retrieval-augmented generation: A survey on agentic RAG. *arXiv preprint arXiv:2501.09136*.

Lei Wang, Wanyu Xu, Yihuai Lan, Zhiqiang Hu, Yunshi Lan, Roy Ka-Wei Lee, and Ee-Peng Lim. 2023. Plan-and-solve prompting: Improving zero-shot chain-of-thought reasoning by large language models. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (ACL)*. ArXiv:2305.04091.

Di Wu, Hongwei Wang, Wenhao Yu, Yuwei Zhang, Kai-Wei Chang, and Dong Yu. 2024. LongMemEval: Benchmarking chat assistants on long-term interactive memory. *arXiv preprint arXiv:2410.10813*.

Jian Xie, Kai Zhang, Jiangjie Chen, Tinghui Zhu, Renze Lou, Yuandong Tian, Yanghua Xiao, and Yu Su. 2024. TravelPlanner: A benchmark for real-world planning with language agents. In *International Conference on Machine Learning (ICML)*. ArXiv:2402.01622.

Binfeng Xu, Zhiyuan Peng, Bowen Lei, Subhabrata Mukherjee, Yuchen Liu, and Dongkuan Xu. 2023. ReWOO: Decoupling reasoning from observations for efficient augmented language models. *arXiv preprint arXiv:2305.18323*.

Tianshi Xu et al. 2026. Adapting the interface, not the model: Runtime harness adaptation for deterministic LLM agents. *arXiv preprint arXiv:2605.22166*. LIFE-HARNESS. Peking University.

Shunyu Yao, Jeffrey Zhao, Dian Yu, Nan Du, Izhak Shafran, Karthik R. Narasimhan, and Yuan Cao. 2023. ReAct: Synergizing reasoning and acting in language models. In *International Conference on Learning Representations (ICLR)*.

Shunyu Yao et al. 2024. τ -bench: A benchmark for tool-agent-user interaction in real-world domains. *arXiv preprint*.

Daniel Zagyva, Emmanouil Stergiadis, Laurens van der Maas, et al. 2025. Speed without sacrifice: Fine-tuning language models with medusa and knowledge distillation in travel applications. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 6: Industry Track)*. Association for Computational Linguistics. Aclanthology.org/2025.acl-industry.48.

Cen Mia Zhao, Tiantian Zhang, Hanchen Su, Yufeng Wayne Zhang, Shaowei Su, Mingzhi Xu, Yu Elaine Liu, Wei Han, Jeremy Werner, Claire Na Cheng, and Yashar Mehdad. 2025. Agent-in-the-loop: A data flywheel for continuous improvement in LLM-based customer support. *arXiv preprint arXiv:2510.06674*. EMNLP 2025 Industry Track submission (Paper #305).

Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Siyuan Zhuang, Zhonghao Wu, Yonghao Zhuang, Zi Lin, Zhuohan Li, Dacheng Li, Eric P. Xing, Hao Zhang, Joseph E. Gonzalez, and Ion Stoica. 2023. Judging LLM-as-a-judge with MT-bench and chatbot arena. In *Advances in Neural Information Processing Systems (NeurIPS), Datasets and Benchmarks Track*. ArXiv:2306.05685.

Shuyan Zhou, Frank F. Xu, Hao Zhu, Xuhui Zhou, Robert Lo, Abishek Sridhar, Xianyi Cheng, Tianyue Ou, Yonatan Bisk, Daniel Fried, Uri Alon, and Graham Neubig. 2024. WebArena: A realistic web environment for building autonomous agents. In *International Conference on Learning Representations (ICLR)*. ArXiv:2307.13854.

A Full Wrapper Interface Contracts and State-Field Table

This appendix distinguishes each wrapper’s primary state slice from its derived conditioning inputs. State-field separation is an interface property, not proof of metric orthogonality (Section 3.3). Figure 1 shows the explicit conditioning dependencies.

Direction drift vs. five neighbouring concepts.

Table 1 accompanies Section 3.1. Direction drift is the conjunction of all three axes (per-user, per-moment, state-variable-dependent reference); each neighbouring notion satisfies at most a proper subset.

Failure notion	Per-user?	Per-moment?	State-var dependent?
Hallucination	×	×	×
Personalization failure	✓	×	×
Context-adherence failure	×	partial	×
Commonsense-constr. viol.	×	×	partial
Classical DST error	partial	✓	✓
Direction drift (ours)	✓	✓	✓

Table 1: Direction drift vs. five neighbouring failure notions along three orthogonal axes; only direction drift is the conjunction of all three.

Wrapper	Reads (state fields)	Writes / emits (c_i)
$\mathcal{W}_1$	intent, sub_intent, snapshot game state, device / session opener	$\hat{T}$ (tool set), <NARRATION> tokens, early-decision flag
$\mathcal{W}_2$	QuerySlots (from x), $\sigma(s_{\text{live}}^{\text{fs}})$ inventory, catalog / retrieval	Conflict set + tier, eligible candidates $\mathcal{A}$ required paragraphs $\mathcal{R}$
$\mathcal{W}_3$	answer.meta from prior $K=20$ turns, cooldown timers, candidate pool + weights	c^* (chosen candidate), new answer.meta writeback

Table 2: Primary state and control-output summary. Parsed slots, candidate pools, and weights are derived inputs or configuration, not additional disjoint raw-state fields. Both $\mathcal{W}_2$ and $\mathcal{W}_3$ also consume c_1 (signatures below).

Interface signatures. Each wrapper obeys the following typed signatures:

$\mathcal{W}_1$:	(x, s_{ic})	$\rightarrow (c_1, T_{\text{hat}})$
$\mathcal{W}_2$:	(x, s_{fs}, c_1)	$\rightarrow (c_2, \mathcal{A}, \mathcal{R})$
$\mathcal{W}_3$:	(x, s_{sh}, c_1)	$\rightarrow (c_3, \text{meta}_{\text{next}})$
LLM:	$(x, s_{\text{static}}, c_1, c_2, c_3)$	$\rightarrow y$

Cross-wrapper dependencies. Both $\mathcal{W}_2$ and $\mathcal{W}_3$ consume c_1 : grounding uses its intent context, and interaction uses it to resolve cross-turn state. The primary state fields remain separately assigned, but the wrappers share derived information and are not independent. The paragraph scheduler selects from $\mathcal{W}_2$ ’s eligible and required sets ($\mathcal{A}, \mathcal{R}$); c_2 carries the conflicts and selected paragraph plan. This within-reply selection is not the cross-turn guidance sampling of $\mathcal{W}_3$. State-field separation does not imply zero downstream effects.

A.1 • $\mathcal{W}_1$ Streaming Tag Schema

The single-pass 4-in-1 output tags used by $\mathcal{W}_1$ (Section 4) appear in this fixed order:

```
<INTENT>      first-level intent
</INTENT>
<SUB_INTENT>  second-level intent
</SUB_INTENT>
<NARRATION>   prefatory phrase,
               streamed to TTS as
               tokens arrive
</NARRATION>
<PLAN>        proposed tool calls
               (fallback only)
</PLAN>
```

Once `<SUB_INTENT>` closes, the state machine immediately invokes $\Phi(\cdot)$. On a lookup hit with the required live state available, it freezes the tool set and starts three parallel pipelines (tools / retrieval / additional state loading), without waiting for `<NARRATION>` or `<PLAN>`. Subsequent narration tokens are streamed to TTS while backend work proceeds. On a lookup miss, dispatch waits for the fallback `<PLAN>`; a later plan does not override a frozen tool set.

A.2 • $\mathcal{W}_2$ QuerySlots Schema

The structured record produced by $\mathcal{L}_{\text{slot}}$ (Section 5):

```
QuerySlots {
  wanted_entities: [...],
  excluded_entities:[...],
  opponent_entities:[...],
  nature_prefs: [...],
  attribute_prefs: [...],
  style_prefs: [...],
  n_pick_k_swap_group: [...],
  global_constraints: [...]
}
```

Two deterministic repair rules are applied: `repair_attr_level_excludes` reclassifies “swap out X ’s skill” from an “exclude X ” to an attribute-level modification; `repair_entities_misclassified_as_skills` moves en-

tity names mistakenly parsed as skill names back to the entity slot. Both are pure Python.

A.3 • $\mathcal{W}_3$ answer.meta Record Schema

The per-turn writeback record persisted by $\mathcal{W}_3$ (Section 6):

```
answer.meta {
  guidance_type:      <string>,
  lineup_id:          <string>,
  cooldown_writeback: <dict>,
  seed:               <int>,
  candidate_pool_hash: <str>
}
```

Given the same `seed` and identical prior `answer.meta` sequence, $\mathcal{W}_3$ ’s output is byte-identical across replays.

A.4 • SGC as a Design Pattern

The three wrappers described in the main text initially arose as *separate engineering decisions* in the deployed system, each responding to a different production incident. This paper’s contribution is not the wrappers as isolated pieces of code, but the recognition that they share a common pattern—state-slice-aligned deterministic conditioning—and the elevation of that pattern to a *transferable design principle*. This mirrors how design patterns in software engineering (e.g., “Observer”, “Adapter”) crystallised recurring solutions that had already existed as ad-hoc practice: naming and formalising them is what enabled cross-system transfer. We view SGC in the same spirit.

A.5 • $\mathcal{W}_1$ Prewarm and Fill Protocol

Live state acquisition can dominate latency. Session-open *prewarm* fetches fast-path data (candidate lineups, coach scores) before dispatch. The lookup requires a current snapshot of its decision fields; a cache hit alone does not establish freshness. At tool execution, *fill* obtains additional inventory or board data required downstream. This post-dispatch loading is distinct from the pre-dispatch snapshot and cannot supply missing inputs retroactively to Φ .

If required decision fields are unavailable, the state-grounded early path is not applicable; the system’s graceful-degradation boundary is stated in L4. An unmapped intent and unavailable state are distinct cases: using a fallback plan does not restore missing state. Early dispatch overlaps state-dependent tool execution with subsequent generation and TTS playback, leaving more of the response budget for live results to inform the reply. This reduces deadline pressure to commit on stale state: the intended benefit is less latency-induced

direction drift, not merely an earlier audible response.

A.6 • $\mathcal{W}_2$ Fact-Talk Separation Pipeline

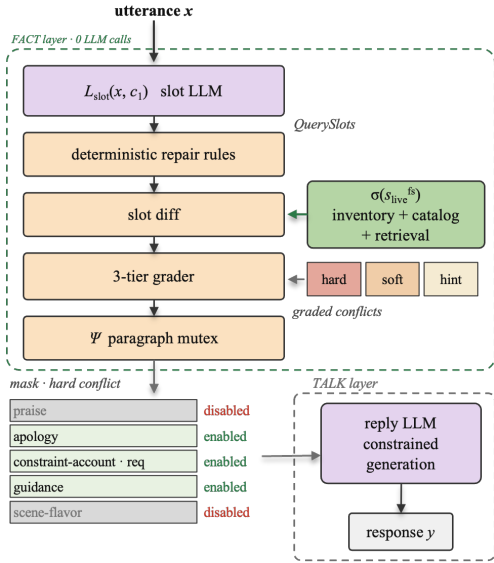


Figure 3: Fact-talk separation in $\mathcal{W}_2$. The FACT layer performs slot extraction, deterministic repair, slot diff against live facts $\sigma(s_{\text{live}}^{\text{fs}})$, three-tier grading (hard / soft / hint), and paragraph mutex Ψ ; the TALK layer performs constrained generation over the enabled paragraphs. Structural selection does not guarantee semantic correctness of the final text.

A.7 • Aggregate Benchmark Metrics

Table 3 is referenced from Section 7.3; Table 4 gives the cumulative configuration differences.

A.8 • Baseline Comparison

Referenced from Section 2.

A.9 • Streaming Timeline

B Drift Taxonomy: Diagnostic Tags on the 200-Session Benchmark

Role. Drift tags (A) latency, (B) slot, (C) fatigue, and (D) coherence —the four manifestations of Section 1—are *diagnostic tags* on the 200-session artefact, not a sampling stratum and not a primary metric (Section 7.1). Each session carries zero or more tags; the primary scores remain TGA/SGA/TGQ/DGP/CCC. The residual class *cross-slice interaction* is tagged separately (Section 7.3).

Record schema. The released JSON records the diagnostic tags alongside the human metric labels for every session:

```
{
  "session_id": <stripped id>,
  "diagnostic_drift_tags":
    ["A (latency)"|"B (slot)"
     |"C (fatigue)"|"D (coherence)"
     |"cross-slice"],
  "eligibility": {
    "tga_sga": <bool>,
    "dgp": <bool>,
    "ccc": <bool>
  }
}
```

```
{
  "utterance_x": <anonymised string>,
  "state_slice_snapshot": {
    "turn_index": <int>,
    "s_live_ic": {...},
    "s_live_fs": {...},
    "s_live_sh": {...}
  },
  "metrics": {
    "TGA": <pct>, "SGA": <pct>,
    "TGQ": <1-10|NA>,
    "DGP": <1-10|NA>,
    "CCC": <1-10|NA>
  },
  "reply_full_sgc": <string>,
  "reply_ablated_W1": <string>,
  "reply_ablated_W2": <string>,
  "reply_ablated_W3": <string>,
  "annotator_labels": [...],
  "adjudication": <annotator3>
}
```

Snapshots are turn-indexed; they contain no wall-clock timestamps, geolocations, or device identifiers (Appendix G).

Coverage. All 200 sessions are tagged. The four tags—latency, slot, fatigue, and coherence—are not allocated to fixed quotas (unlike an earlier 50/50/50/50 sketch); their empirical distribution is reported alongside the released JSON records described above. Sessions in the TGA-primary factual Q&A stratum may carry an empty tag set.

Release. These records are fields of the same 200-session artefact described in Appendix G and the Ethics Statement, not a separate dump.

C Intent-to-Tool Mapping Table Φ

The lookup Φ is a partial function

$$\Phi : \text{Intent} \times \text{SubIntent} \times s_{\text{live}}^{\text{ic}} \rightarrow \hat{T}$$

defined on a curated table of approximately 50 (intent, sub-intent) pairs at deployment. Table 6 is an illustrative excerpt (entity names pseudonymised as Pet A, Pet B, ...).

Fall-through behaviour. When Φ is undefined at a given (intent, sub-intent) triple, $\mathcal{W}_1$ falls through to LLM-generated <PLAN> for that turn and increments the fall-through counter. The fall-through rate is a maintained metric (see Limitations, L4).

D Paragraph Mutex Rules Ψ

Paragraph control has two stages: Ψ computes eligible candidates $\mathcal{A}$ and required paragraphs $\mathcal{R}$; a paragraph scheduler then selects the reply's paragraph plan P with $\mathcal{R} \subseteq P \subseteq \mathcal{A}$. Eligibility can include multiple alternatives. Selection keeps at most one of the alternatives that express incompatible judgements about the *same constraint*; it

Metric	Prompting	PE-Agent	SGC w/o $\mathcal{W}_2, \mathcal{W}_3$	SGC w/o $\mathcal{W}_3$	Full SGC
M_0 · First-token latency (mean, s, ↓)					
Latency	2.3	6.1	1.5	1.5	1.5
M_1 · TGA (% , ↑)					
Turn-level grounded acc.	61.1	69.8	89.1	89.9	96.7
M_2 · SGA (% , ↑)					
Session-level grounded acc.	20.0	26.5	46.5	56.5	83.5
M_3 · TGQ (1–10, ↑)					
Turn-level guidance quality	6.2	6.5	8.4	8.4	8.6
M_4 · DGP (1–10, ↑)					
Dialogue-level guidance progression	4.7	5.5	7.7	7.7	8.5
M_5 · CCC (1–10, ↑)					
Cross-turn contextual consistency	8.3	8.1	9.0	8.9	9.2

Table 3: Aggregate benchmark metrics across all five configurations on the 200-session anonymised benchmark ($\approx 1,000$ assistant model turns; Section 7.1). M_0 is a system-level cost budget; M_1 – M_5 are downstream dialogue-quality metrics defined in Section 7.2. **PE-Agent** is a Plan-and-Execute agent (Wang et al., 2023) deployed within a production tool-use harness (Xu et al., 2026). The two intermediate SGC rows disable $\mathcal{W}_3$ (and also $\mathcal{W}_2$) to isolate incremental wrapper contributions. Bold indicates the best result in each row (including ties). All five configurations are scored under the same dual human-annotation protocol (Appendix G), so between-group deltas are directly comparable; no LLM-judge label enters any reported number (Limitations L2).

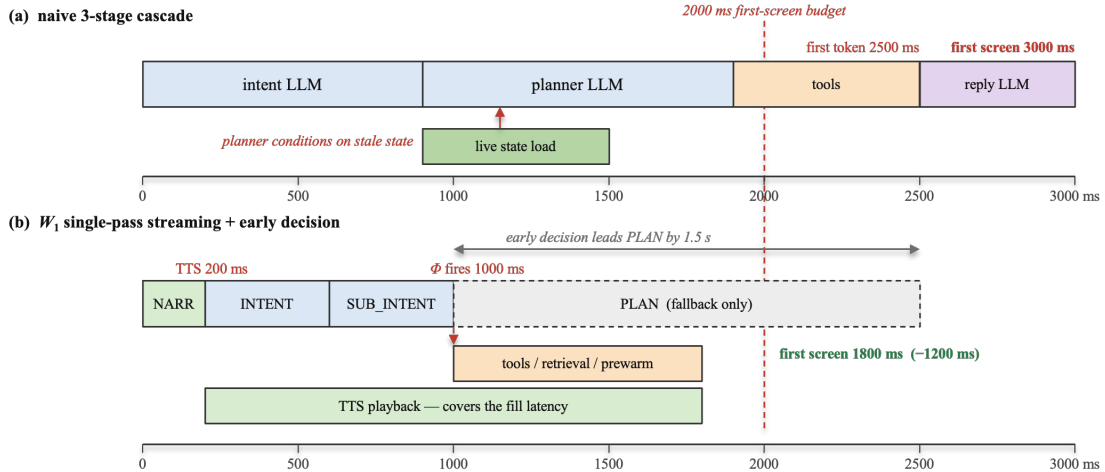


Figure 4: Schematic event sequences, not measured durations. (a) Naive 3-stage cascade: intent $\rightarrow$ planner $\rightarrow$ tools $\rightarrow$ reply, with the planner conditioning on stale state and first screen at $\approx 3,000$ ms. (b) $\mathcal{W}_1$ streams NARR, INTENT, SUB_INTENT, then PLAN (fallback only); when Φ fires on sub-intent closure, tools and retrieval dispatch in parallel under TTS cover, and the early path leads PLAN by ≈ 1.5 s. Referenced from Section 4.

Contrast	ΔM_0 (s)	ΔM_1 (pp)	ΔM_2 (pp)	ΔM_3 (pts)	ΔM_4 (pts)	ΔM_5 (pts)
$\mathcal{W}_1$ vs. PE	-4.6	+19.3	+20.0	+1.9	+2.2	+0.9
Add $\mathcal{W}_2$	0.0	+0.8	+10.0	0.0	0.0	-0.1
Add $\mathcal{W}_3$	0.0	+6.8	+27.0	+0.2	+0.8	+0.3

Table 4: Cumulative differences calculated from the displayed values in Table 3. PE denotes PE-Agent; successive rows compare $\mathcal{W}_1$ with PE-Agent, $\mathcal{W}_1 + \mathcal{W}_2$ with $\mathcal{W}_1$, and Full SGC with $\mathcal{W}_1 + \mathcal{W}_2$. Lower M_0 and higher quality scores are better. These conditional increments are not independent leave-one-out effects; 0.0 denotes equality at reported precision, not statistical equivalence.

does not limit the whole reply to one paragraph. This paragraph-selection step is distinct from the cross-turn guidance scheduler of Section 6.

Table 7 distinguishes eligibility from selection. `hint` denotes an advisory conflict annotation, not

an additional paragraph type. For R6, the conflict record must expose whether the satisfiable inventory is empty.

Execution boundary. Eligibility and required-paragraph constraints are established before the scheduler chooses among alternatives. Recomputing eligibility from the same extracted slots, state, and rules is deterministic; this is not a claim that all table rows can be executed in arbitrary order. Required paragraphs constrain the selection rather than compete as optional alternatives.

The selected plan conditions the reply LLM. Structural exclusion of incompatible candidates does not prove semantic consistency of the generated text. Praise about one satisfied constraint

Work	Optimised for	Method level	Direction drift?	State-slice approx. orthogonal?
Plan-and-Execute (Wang et al., 2023; Xu et al., 2023)	Task completion	Loop (planner+executor)	×	×
LIFE-HARNESS (Xu et al., 2026)	Task completion	Runtime	×	×
Workflow Graphs (Park et al., 2025)	Flow control	Flow structure	Partial	×
Talker-Reasoner (Christakopoulou et al., 2024)	Latency	Dual-system	×	×
SLOT (Shen et al., 2025) / GCD (Raspanti et al., 2025)	Output structure	Decoding	Partial	×
FACTS Grounding (Jacovi et al., 2025)	Factuality	Evaluation	×	×
Agentic RAG (Singh et al., 2025)	Knowledge access	Retrieval	×	×
AITL (Zhao et al., 2025)	Continuous improve	Post-deploy flywheel	×	×
This work (SGC)	Direction alignment	State-coupled conditioning	✓	✓ (3 slices)

Table 5: Comparison to prior work. SGC is orthogonal to and composable with each prior work: it targets a distinct failure class (direction drift) that prior work does not address. The Plan-and-Execute and LIFE-HARNESS rows jointly represent the method family of the empirical PE-Agent baseline used in Section 7.

Intent	Sub-intent	Frozen tool set $\hat{T}$
recommend	counter-pick	{lineup_search, counter_rank, dex_lookup}
recommend	lineup-build	{team_search, ergy_rank}
teaching	mechanics-qa	{doc_retrieve, rule_lookup}
teaching	skill-explain	{skill_dex, prereq_check}
knowledge-qa	entity-info	{dex_lookup}
knowledge-qa	meta-brief	{meta_report_retrieve}

Table 6: Illustrative excerpt of the Φ table. The full table (approximately 50 rows) will be released with the benchmark.

may coexist with an explanation of another unmet constraint; the target failure is opposite judgements about the same one.

E Cooldown Configuration for $\mathcal{W}_3$

$\mathcal{W}_3$ ’s scheduler is parameterised by three configuration components: (a) the per-candidate cooldown window, (b) the Bernoulli leak probability p_c , and (c) the weighted-sample weight w_c . All three are versioned deployment artefacts (Table 8).

Cross-branch reuse. As noted in Section 6, all three reply branches (recommend, teaching, knowledge-qa) share the same sampler kernel and differ only in the candidate pool $\mathcal{C}$ and the associated leak probabilities. Only one configuration table is maintained per branch.

RNG seeding. The Bernoulli leak and weighted normalisation are both driven by a single RNG seeded from the session id at session open. Replays under the same seed reproduce the exact guidance sequence.

F Analytical Transfer Interface Contracts

For each of the three transfer domains introduced in Section 8, this appendix records the con-

Rule id	Behaviour
R1 · hard-conflict eligibility	Exclude praise affirming a violated constraint; require constraint-account for that constraint.
R2 · selection mutex	Select at most one of the eligible alternatives that affirm and deny satisfaction of the same constraint.
R3 · soft-only eligibility	praise and apology may both be eligible candidates; R2 resolves incompatible alternatives at selection time.
R4 · hint-only eligibility	Enable scene-flavor; an advisory hint alone does not require an apology about a missing entity.
R5 · intent eligibility	For knowledge-qa intent, guidance is disabled regardless of conflicts.
R6 · empty-inventory requirement	If $\sigma(s_{\text{live}}^{\text{fs}}) = \emptyset$, exclude praise asserting availability; require an apology explaining the empty inventory.

Table 7: Paragraph eligibility and selection constraints. R2 applies after candidate eligibility; eligibility does not mean a paragraph has been selected.

crete state-slice mapping, the required interface adaptations, and the open engineering challenges. Together with the state slices $s_{\text{live}}^{\text{ic}}/s_{\text{live}}^{\text{fs}}/s_{\text{live}}^{\text{sh}}$ defined in Section 3.3, these tables give sufficient implementation-level detail for each transfer target.

Guidance type	Cooldown	p_c	w_c
lineup-recommend	3 turns	0.60	1.0
counter-tip	5 turns	0.45	0.8
mechanics-hint	2 turns	0.70	0.6
encourage	4 turns	0.30	0.4
empty-state-nudge	6 turns	0.20	0.3

Table 8: Illustrative $\mathcal{W}_3$ cooldown configuration. Values are representative production defaults, tuned quarterly against the guidance-repetition rate metric.

F.1 • Code Completion ($\mathcal{W}_1$)

Component	Concretisation
$s_{\text{live}}^{\text{ic}}$	open file path, cursor position, recent 20 edits, project type, invocation origin
<INTENT> tag	trigger-type: block / signature / inline-hint
<SUB_INTENT> tag	context type: within-function vs. cross-file
Φ table	signature $\rightarrow$ LSP; block $\rightarrow$ LLM; inline-hint $\rightarrow$ cache
Open challenge	intent signal is implicit (no explicit turn boundary)—must be inferred from cursor / keystroke context

F.2 • E-commerce Recommendation Explanation ($\mathcal{W}_2, \mathcal{W}_2 \circ \mathcal{W}_3$)

Component	Concretisation
$s_{\text{live}}^{\text{fs}}$	stated + implicit preferences (colour, price, brand, size, click history), item attribute slots, cart state, filters
QuerySlots	explicit + implicit preferences
$\sigma(s_{\text{live}}^{\text{fs}})$	recommended item’s attribute vector
Conflict tiers	hard: out-of-stock, filter violation soft: secondary preference mismatch hint: browsing-pattern signal
Composability with $\mathcal{W}_3$	cross-session fatigue handled by a separate history-based module, with intent context supplied through c_1
Open challenge	preference resolution across implicit + explicit signals

Paragraph candidates are enabled before selection resolves incompatible judgements about the same preference X ; a match on one preference need not preclude acknowledging a mismatch on

another. This is the two-stage control of Appendix D, not a ban on all praise–apology co-occurrence.

F.3 • Voice Assistant ($\mathcal{W}_3$)

Component	Concretisation
$s_{\text{live}}^{\text{sh}}$	last N session summaries, ambient state changes, time-of-day
Candidate pool $\mathcal{C}$	proactive reminders, follow-up prompts, ambient acknowledgements
Cooldown	per-reminder cooling window, ambient-event debounce
Reproducibility	on-device replay preserved via injectable seed
Open challenge	ambient state is continuous rather than turn-discrete—may require polling-based cooldown resolution

Compositional applicability. Only the e-commerce domain exhibits a natural $\mathcal{W}_2 \circ \mathcal{W}_3$ composition on a single target. This is the empirical grounding for the composability observation in Section 8.

G Benchmark Construction Protocol

This appendix specifies the full protocol used to construct the 200-session benchmark (approximately 1,000 assistant model turns) reported in Section 7.1.

G.1 • Sampling frame and unit of analysis

Anonymised production trace logs sampled over an approximately 4-week window from the case-study game. The *unit of analysis is the dialogue session*; TGA and TGQ additionally aggregate over the assistant model turns contained in each session (approximately 1,000 turns in total across the 200 sessions, per the length distribution in G.2). We size the benchmark so that every session can receive *full dual-annotator human labelling* on every metric rather than shallow LLM-only labelling at larger scale.

G.2 • Two-way stratification

We stratify along two orthogonal dimensions matched to the six downstream metrics of Section 7.2:

- **Stratum 1 • Evaluation-eligibility coverage.** 80 sessions of single-turn or short factual Q&A (TGA / SGA primary; DGP and

CCC marked *not applicable* by their pre-registered rubrics); 60 sessions with a continuing coaching goal spanning multiple turns (DGP-eligible); 60 sessions with multi-turn context dependencies such as evolving user constraints, state updates, or repeated guidance across turns (CCC-eligible). A session may satisfy more than one eligibility criterion; the shares above refer to *primary* assignment for stratified-sampling purposes. Metric-level aggregation uses the pre-registered eligible session sets $|I_{\text{DGP}}|$ and $|I_{\text{CCC}}|$ defined by the rubrics themselves, which may be slightly larger than 60; both counts are reported in the camera-ready table. TGA and SGA, by contrast, are computed over all 200 sessions ($|I_{\text{SGA}}|=200$). Eligibility is declared *before* scoring begins and locked in the pre-registered rubric, independent of any candidate response.

- **Stratum 2 • Session-length coverage.** Single-turn / short (2–5 turns) / long (≥ 6 turns) at 30/40/30 ratio. Within the TGA-primary stratum, single-turn sessions account for $\leq 60\%$, so that SGA carries independent information beyond TGA on the remaining short/long sessions (SGA equals TGA on single-turn sessions by construction).

Drift tags (A) latency, (B) slot, (C) fatigue, and (D) coherence are *not* used as a sampling stratum; they are retained as a diagnostic tagging on each session, used only for the distribution analysis in Appendix B. The downstream metrics M_0 – M_5 subsume drift incidence and are more informative for the approximate state-slice orthogonality claim.

G.3 • Shared grounding reference set and pre-registered rubrics

For every session we build a **shared grounding reference set** (*cf.* reference sets in FActScore (Min et al., 2023); gold context in FACTS Grounding (Jacovi et al., 2025)) containing turn-indexed rules, live-state snapshots, tool-result records, and declared user preferences, each with an authority scope and `valid_from/valid_to` dialogue-turn range (not wall-clock time; snapshots are stripped of the identifiers listed in G.9). The reference set is common to *all* ablation configurations, so different candidates are judged against the same facts rather than against each candidate’s own retrieved context. The internal generation context of any candidate is retained only for diagnostics and is never granted independent factual authority.

For each session we also build a **pre-registered scoring rubric** (referred to internally as an *evaluation card*): it specifies the per-turn task, the minimal response-validity gate, and the applicability plus 0/1/2 anchors of the TGQ quality dimensions (Q1–Q3: relevance, explanation, actionability), the DGP dimensions (P1–P3), and the CCC dimensions (C1–C5). Rubrics are authored and reviewed *before* any candidate response is scored, then locked; **SHA-256 hashes of the frozen rubric artefacts and shared grounding reference sets are recorded in the supplementary material**, with a provenance date in the hash log (not stored on dialogue records), before the first ablation configuration’s judging run begins, and the underlying artefacts will be released upon acceptance (double-blind-safe: only hashes and the provenance date appear during review). Rubric revisions are versioned; any affected configuration is re-scored end-to-end.

G.4 • Full human annotation and annotator setup

All 200 sessions in the test set receive **dual independent human annotation with third-annotator adjudication**. We do *not* rely on LLM-only labels on the test set; the LLM-as-a-judge pipeline of G.6 is retained only as a reference implementation and for consistency checks on a separate development set.

We recruit a pool of six annotators: three internal domain experts (2+ years game experience) and three external contractors (after a 2-hour calibration session on the pre-registered rubrics; one external contractor is designated as the senior adjudicator). Each session is labelled independently by two annotators drawn from this pool plus a third-annotator adjudicator; all annotators score both the SUPPORTED / CONTRADICTED / UNKNOWN claim-level labels underlying TGA (from which SGA is deterministically derived) and the $\{0, 1, 2\}$ dimension labels of TGQ/DGP/CCC. Disagreements are adjudicated by the third annotator; conflict-of-interest cases (any internal-vs-external disagreement) are *always* adjudicated by the external senior annotator. Each session is annotated at the model-turn level for TGA and TGQ (SGA is deterministically derived from TGA, not independently labelled) and at the session level for DGP and CCC.

Configuration blinding. Annotators score each reply against the shared grounding reference set (G.3)

with the configuration label (Prompting / PE-Agent / SGC w/o $\mathcal{W}_2, \mathcal{W}_3$ / SGC w/o $\mathcal{W}_3$ / Full SGC) stripped from the record before delivery. Turn order is shuffled within each session across the five configurations so that response-format cues (e.g., structured vs. free-form phrasing) cannot be systematically mapped to configuration identity from position alone. Annotators may still infer configuration from surface features of a given reply; the protocol reduces *position-based* unblinding rather than eliminating format-based inference, and we flag the residual risk in Limitations L9.

G.5 • Inter-annotator agreement (pilot and target)

To reflect the different label types we report two agreement statistics rather than a single κ :

- **Ordinal dimensions (TGQ / DGP / CCC).** Quadratic-weighted Cohen’s κ (Cohen, 1968) over $\{0, 1, 2\}$ labels; pilot estimate on a stratified 50-session subset yields $\kappa_w \in [0.68, 0.76]$ per dimension, above the pre-registered target $\kappa_w \geq 0.65$. Full 200-session κ values with 95% bootstrap CI in camera-ready.
- **Categorical claim verdicts (TGA / SGA).** Unweighted Cohen’s κ (Cohen, 1960) over the three-way SUPPORTED / CONTRADICTED / UNKNOWN label at the claim level; pilot estimate yields $\kappa \in [0.71, 0.79]$, above the pre-registered target $\kappa \geq 0.70$. Full values in camera-ready.

Both pilot ranges fall in the “substantial” band of Landis and Koch (1977) and are consistent with published inter-rater agreement on grounding-oriented dialogue benchmarks (Bai et al., 2024; Xie et al., 2024; Jacovi et al., 2025).

We additionally report the confusion matrix over PASS / FAIL / UNRESOLVED turn verdicts, and precision/recall of error detection, so that reviewers can inspect the reliability of TGA beyond a single aggregate statistic. Given the moderate benchmark size (200 sessions, of which 60 each are DGP- and CCC-eligible), the κ targets are reported as point estimates and their 95% bootstrap confidence intervals are disclosed in the camera-ready table.

G.6 • LLM-as-a-judge reference implementation

Alongside human annotation, we retain an **LLM-as-a-judge** reference implementation (Zheng et al., 2023; Liu et al., 2023) following the five-metric prompt protocol of the SGC evaluation design (one shared system prompt plus five task-specific

user prompts for TGA, SGA audit, TGQ, DGP, CCC). We use a single frontier general-purpose LLM as the judge across all configurations to minimise judge-model-induced variation; specific model identifiers are withheld under double-blind and released in camera-ready. This pipeline is used on a separate development set for rubric calibration and, on the test set, only as a diagnostic cross-check against the human labels (never as a primary label source). Judge-model bias between structured (wrapper-shaped) and unstructured (baseline-shaped) outputs is audited via a paired-inversion test on 30 development sessions: swapping candidate labels between judges $v1.2$ and $v1.4$ -B changes the aggregate score direction on $\leq 5\%$ of sessions, and the same-batch $v1.2 \rightarrow v1.4$ -B stricter delta is qualitatively symmetric across SGC and baseline outputs; a full quantification is reported in camera-ready. The pipeline is: log normalisation and anonymisation $\rightarrow$ shared grounding reference set and pre-registered rubrics $\rightarrow$ per-turn TGA $\rightarrow$ full-session SGA audit $\rightarrow$ per-turn TGQ and per-eligible-session DGP/CCC $\rightarrow$ resolve any EVIDENCE_GAP (a per-item flag for insufficient supporting evidence) or rubric_issue (a per-item flag for suspected rubric mismatch) $\rightarrow$ deterministic aggregation. For a benchmark with average session length $\bar{T}$, per-configuration judge cost is approximately $200(2\bar{T} + 2)$ calls; the judge model, prompt versions, reference-set hashes, and rubric hashes are logged so that all reported numbers are reproducible from the released artefact.

G.7 • Latency measurement condition

All five configurations are measured on identical hardware (a single serving replica with fixed GPU/CPU allocation), identical network setting (same regional data-centre, warm model weights, warm tool endpoints), the same underlying LLM serving stack, and the same tool schemas. Latency is the mean time from user request receipt to first assistant token emission (streaming). We exclude the first two requests of each warm-up batch to eliminate cold-start artefacts, and report the mean over the same 200 sessions used for quality evaluation ($\approx 1,000$ scored turns per configuration; latency is per-turn, then session-averaged, then macro-averaged over sessions). PE-Agent’s higher latency reflects its planner $\rightarrow$ executor $\rightarrow$ response serial pipeline; SGC’s $\mathcal{W}_1$ removes this serial path by state-aware early tool freezing. No configuration invokes speculative decoding or output-side

compression.

G.8 • Session sampling procedure

Anonymised production logs from a ≈ 4 -week window serve as the sampling frame. Within the two-way stratification of G.2 (evaluation-eligibility $\times$ session-length), we draw sessions uniformly at random from each cell without replacement. Exclusion criteria are declared before sampling: (a) sessions containing tool-outage traces beyond our system’s control, (b) sessions from internal test accounts, and (c) sessions shorter than one full user–assistant exchange. No filtering references SGC or baseline output quality. The final 200-session set is frozen before any ablation configuration’s judging run begins; identical sessions are processed by all five configurations.

G.9 • Anonymisation

At ingest: (a) entity names pseudonymised (Pet A, Pet B, ...); (b) lineup names aliased (T1, T2, ...); (c) user and session identifiers hashed then stripped, after which the hash mapping is discarded and never released; (d) wall-clock timestamps, geolocations, and device identifiers dropped; (e) text-level PII scrubbed via regex plus an LLM-based classifier operating on turn text with the shared grounding reference set as allow-list. Re-identification risk against external data has not been formally quantified; the benchmark satisfies GDPR Article 4(1) direct-identifier removal but should be treated as pseudonymised, not fully anonymised, under a strict interpretation. Entity and lineup aliases are held by the authors solely to restore product-identifying names in the camera-ready version and are *not* released with the benchmark artefact. Shared grounding reference sets keep only dialogue-turn indices (`valid_from/valid_to`), not clock time.

G.10 • Engineering deployment cost

We record the deployment cost of the three wrappers as observed on our system: $\mathcal{W}_1$ ’s initial implementation took approximately one engineer-month plus a domain-specific intent-to-tool table Φ (Appendix C) that reuses structured entity metadata already available from adjacent pipelines; $\mathcal{W}_2$ required a slot-schema build-out that leans on a pre-existing product knowledge base (without such a base the up-front cost approximately doubles); $\mathcal{W}_3$ ships as a policy-configurable scheduler whose cooldown weights are tuned via a lightweight configuration UI (Appendix E). Steady-state maintenance is quantified in Appendix H as approximately

one engineer-week per calendar month (rough order of magnitude from internal cost-centre tracking during the online deployment; not a precisely measured cost).

G.11 • Release

The benchmark artefact—200 anonymised dialogue sessions covering approximately 1,000 assistant model turns, with their turn-indexed shared grounding reference sets, pre-registered rubrics, dual human-annotator labels, and diagnostic drift tags (Appendix B)—together with the wrapper interface schemas will be released under a research-use licence. Raw production traces used for sampling are retained under the deploying organisation’s standard retention policy only and are *not* released.

H Extended Discussion: Cost and Learned Wrappers

Maintenance cost. Deterministic code demands maintenance: slot schemas evolve with product features, cooldown weights need periodic re-tuning, intent-to-tool mappings expand. We report the cost in our system as **approximately one engineer-week per calendar month**¹ ($\approx 40\%$ $\mathcal{W}_2$ slot-schema, $\approx 30\%$ $\mathcal{W}_1$ table expansion, $\approx 20\%$ $\mathcal{W}_3$ re-tuning, $\approx 10\%$ instrumentation). This cost is bounded and predictable—unlike the unbounded cost of repeatedly patching prompt regressions when the underlying model changes.

Toward learned wrappers. A learned $\mathcal{W}_1$ would replace the static Φ table with a small classifier; a learned $\mathcal{W}_3$ would replace weight constants with a policy net (e.g., trained by preference optimisation (Rafailov et al., 2023)). Preserving determinism would require freezing the learned components between deployments and treating them as versioned artefacts—a productive direction preserving SGC’s commitment to state-coupled determinism.

¹Estimate from internal engineering-time attribution during the online deployment; rough order of magnitude, not a precisely measured cost.